

Flight Booking System - Full API Contract

API Access Classification

APIs in the Flight Booking System can be classified based on who is allowed to access them.

1. Public Endpoints

- Public endpoints can be accessed without login.
- Public endpoints do not require a JWT token.
- These endpoints are used for actions that must be available before authentication, such as customer registration, login, flight search, and viewing public flight information.

Public endpoints in this system include:

- POST /api/auth/register
 - Allows a new customer to create an account.
- POST /api/auth/login
 - Allows users to log in and receive a JWT access token.
- GET /api/airports
 - Returns the list of airports used in flight search.
- GET /api/airports/{airportId}
 - Returns the details of a specific airport.
- GET /api/flights/search
 - Allows users to search for available flights.
- GET /api/flights/{flightId}
 - Returns the details of a specific flight.
- GET /api/flights/{flightId}/seats
 - Returns the seats of a specific flight.
- GET /api/flights/{flightId}/seats/available
 - Returns the available seats for a specific flight.

2. Protected Endpoints

- Protected endpoints require authentication.
- The user must be logged in.
- The frontend must send the JWT access token in the request header.

Authorization: Bearer <access_token>

- Protected endpoints are also controlled using role-based access control.

- The main roles in this system are:
 - CUSTOMER
 - AIRLINE_ADMIN
 - SUPER_ADMIN

3. Auth and Profile Protected Endpoints

These endpoints require a valid JWT token, but they are not limited to one specific role. Any logged-in user can use them to manage their own account.

- GET /api/auth/me
 - Returns the current logged-in user profile.
- PUT /api/auth/me
 - Updates the current logged-in user profile.
- PUT /api/auth/change-password
 - Allows the current logged-in user to change their password.

4. Customer Protected Endpoints

These endpoints require authentication and can be accessed by users with the CUSTOMER role.

- POST /api/bookings
 - Creates a new booking for a selected flight.
- POST /api/bookings/{bookingId}/passengers
 - Adds passenger details and selected seats to a booking.
- GET /api/bookings/my
 - Returns the bookings of the logged-in customer.
- GET /api/bookings/{bookingId}
 - Returns the details of a booking.
 - Access may be allowed for CUSTOMER, AIRLINE_ADMIN, or SUPER_ADMIN depending on ownership and role.
- PATCH /api/bookings/{bookingId}/cancel
 - Allows a customer to cancel an eligible booking.
- POST /api/payments
 - Creates a payment request for a booking.
- GET /api/payments/{paymentId}
 - Returns payment details.
 - Access may be allowed for CUSTOMER or SUPER_ADMIN.
- GET /api/bookings/{bookingId}/payment
 - Returns the payment information related to a booking.
- GET /api/payments/my

- Returns the payments of the logged-in customer.
- POST /api/refunds
 - Allows a customer to request a refund for an eligible paid and cancelled booking.
- GET /api/refunds/{refundId}
 - Returns refund details.
- GET /api/bookings/{bookingId}/refund
 - Returns refund information related to a booking.
- GET /api/refunds/my
 - Returns the refund requests of the logged-in customer.
- GET /api/tickets/my
 - Returns the tickets of the logged-in customer.
- GET /api/tickets/{ticketId}
 - Returns the details of a specific ticket.
- GET /api/bookings/{bookingId}/tickets
 - Returns all tickets related to a specific booking.
- GET /api/notifications/my
 - Returns the notifications of the logged-in customer.
- GET /api/notifications/{notificationId}
 - Returns the details of a specific notification.

Customers must only access their own bookings, payments, refunds, tickets, and notifications.

5. Airline Admin Protected Endpoints

These endpoints require authentication and can be accessed by users with the AIRLINE_ADMIN role.

- POST /api/flights
 - Creates a new flight for the airline assigned to the logged-in Airline Admin.
- PUT /api/flights/{flightId}
 - Updates flight information for the assigned airline.
- PATCH /api/flights/{flightId}/cancel
 - Cancels a flight that belongs to the assigned airline.
- GET /api/flights
 - Returns flights.
 - Airline Admins can view only flights related to their assigned airline.
- POST /api/flights/{flightId}/seats
 - Creates seats for a specific flight.
- PUT /api/seats/{seatId}

- Updates seat information.
- GET /api/bookings/airline
 - Returns bookings related to the assigned airline.
- GET /api/reports/bookings
 - Returns booking reports.
 - Airline Admins can only view reports related to their assigned airline.

Airline Admins must not manage flights, seats, bookings, or passengers that belong to another airline.

6. Super Admin Protected Endpoints

These endpoints require authentication and can be accessed only by users with the SUPER_ADMIN role.

- GET /api/users/customers
 - Returns all customer accounts.
- GET /api/users/customers/{customerId}
 - Returns the details of a specific customer.
- PATCH /api/users/customers/{customerId}/deactivate
 - Deactivates a customer account.
- POST /api/airlines
 - Creates a new airline.
- PUT /api/airlines/{airlineId}
 - Updates airline information.
- PATCH /api/airlines/{airlineId}/deactivate
 - Deactivates an airline.
- GET /api/airlines
 - Returns all airlines.
- POST /api/airline-admins
 - Creates a new Airline Admin account.
- PUT /api/airline-admins/{adminId}
 - Updates an Airline Admin account.
- PATCH /api/airline-admins/{adminId}/deactivate
 - Deactivates an Airline Admin account.
- GET /api/airline-admins
 - Returns all Airline Admin accounts.
- POST /api/airports
 - Creates a new airport.
- PUT /api/airports/{airportId}
 - Updates airport information.
- GET /api/bookings

- Returns all bookings across the system.
- GET /api/refunds
 - Returns all refund requests.
- GET /api/notifications
 - Returns all notifications.
- POST /api/notifications/{notificationId}/retry
 - Retries a failed notification.
- GET /api/reports/payments
 - Returns payment reports.
- GET /api/reports/refunds
 - Returns refund reports.
- GET /api/audit-logs
 - Returns audit logs.
- GET /api/audit-logs/{auditLogId}
 - Returns the details of a specific audit log.

Super Admin operations must be restricted to authorized Super Admin accounts only.

7. Special Protected Endpoints

Some endpoints are protected, but they are not accessed by normal users through the frontend.

- POST /api/payments/callback
 - Access: PAYMENT_GATEWAY
 - Used by the payment gateway to send the payment result to the backend.
 - Protected using a gateway signature header.

X-Gateway-Signature: <gateway_signature>

- POST /api/refunds/callback
 - Access: PAYMENT_GATEWAY
 - Used by the payment gateway to send the refund result to the backend.
 - Protected using a gateway signature header.
- POST /api/bookings/{bookingId}/tickets/generate
 - Access: SYSTEM_INTERNAL
 - Used internally by the system after successful payment confirmation.
- POST /api/notifications/send
 - Access: SYSTEM_INTERNAL
 - Used internally by the system to send booking, ticket, cancellation,

refund, or flight update notifications.

- PATCH /api/tickets/{ticketId}/cancel
 - Access: SYSTEM_INTERNAL or SUPER_ADMIN
 - Used to cancel or invalidate a ticket when the related booking is cancelled.

Payment gateway callbacks must be verified before updating payment, booking, or refund statuses.

12. API Contract

1. General API Rules

- All endpoints return and receive data in JSON format.
 - **Base URL:** /api
 - All protected endpoints require authentication using a JWT token sent in the Authorization header: Authorization: Bearer <access_token>
 - The system applies role-based access control to restrict access based on the user role: CUSTOMER, AIRLINE_ADMIN, SUPER_ADMIN
 - **Public endpoints** can be accessed without login, such as customer registration, login, flight search, and flight details.
 - **Protected endpoints** require login, such as creating bookings, making payments, viewing tickets, managing flights, managing airlines, viewing reports, and audit logs.
 - All dates and times should use ISO format: 2026-07-01T14:30:00
 - The system should not permanently delete important historical records such as bookings, payments, tickets, refunds, and audit logs. Deactivation or status updates should be used instead of deletion where needed.
-

2. Standard Response Format

handle successful responses, errors, validation messages, and paginated data in a consistent way.

2.1 Success Response Format

```
{ "success": true,  
  "message": "Operation completed successfully",  
  "data": {}  
}
```

2.2 Error Response Format

```
{  
  "success": false,  
  "message": "Something went wrong",  
  "errorCode": "ERROR_CODE",  
  "details": []  
}
```

2.3 Validation Error Response Format

```
{  
  "success": false,  
  "message": "Validation failed",  
  "errorCode": "VALIDATION_ERROR",  
  "details": [  
    "Email is required",  
    "Password must be at least 8 characters"  
  ] }  
}
```

2.4 Paginated Response Format

```
{  
  "success": true,  
  "message": "Data retrieved successfully",  
  "data": {  
    "content": [],  
    "page": 0,  
    "size": 10,  
    "totalElements": 100,  
    "totalPages": 10  
  }  
}
```

3. Auth APIs

- Who can register?
- How do they log in?
- How do they get a token?
- How do they send the token afterward?
- How can they view or update their profile?
- How can they change their password?
- Customer accounts can be created through public registration. However, Airline Admin accounts cannot register directly from the public registration endpoint. Airline Admin accounts are created only by the Super Admin.
- After successful login, the backend returns a JWT access token. The frontend must send this token with every protected request using the following header: Authorization: Bearer <access_token>
- Public Auth endpoints can be accessed without a token, such as customer registration and login. Protected Auth endpoints require a valid JWT token, such as viewing the current user profile, updating profile information, and changing password.

3.1 Register Customer Creates a new customer account.

Method: POST

Endpoint: /api/auth/register

Access: Public

Request Body:

```
{
  "firstName": "Ahmed",
  "lastName": "Ali",
  "email": "ahmed@example.com",
  "password": "Password@123",
  "phone": "+201001234567"
}
```

Success Response:

```
{
  "success": true,
  "message": "Customer registered successfully",
  "data": {
    "userId": 1,
    "firstName": "Ahmed",
    "lastName": "Ali",
    "email": "ahmed@example.com",
    "phone": "+201001234567",
    "role": "CUSTOMER",
    "userStatus": "ACTIVE"
  }
}
```

Failure Responses:

400 Bad Request

409 Email Already Exists

3.2 Login Authenticates the user using email and password. If the credentials are valid, the backend returns a JWT access token.

Method: POST

Endpoint: /api/auth/login

Access: Public

Request Body:

```
{
  "email": "ahmed@example.com",
  "password": "Password@123"
}
```

Success Response:

```
{
  "success": true,
  "message": "Login successful",
  "data": {
    "accessToken": "jwt-access-token",
    "tokenType": "Bearer",
  }
}
```

```
"user": {
  "userId": 1,
  "firstName": "Ahmed",
  "lastName": "Ali",
  "email": "ahmed@example.com",
  "role": "CUSTOMER",
  "userStatus": "ACTIVE",
  "airlineId": null
}}
```

Failure Responses:

401 Invalid Email or Password

403 Account Deactivated

3.3 Get Current User Profile Returns the profile information of the currently logged-in user.

Method: GET

Endpoint: /api/auth/me

Access: Protected

Headers: Authorization: Bearer <access_token>

Success Response:

```
{ "success": true,
  "message": "User profile retrieved successfully",
  "data": {
    "userId": 1,
    "firstName": "Ahmed",
    "lastName": "Ali",
    "email": "ahmed@example.com",
    "phone": "+201001234567",
    "role": "CUSTOMER",
    "userStatus": "ACTIVE",
    "airlineId": null
  }
}
```

Failure Responses:

401 Unauthorized

403 Forbidden

3.4 Update Current User Profile Allows the logged-in user to update their personal profile information.

Method: PUT OR PATCH

Endpoint: /api/auth/me

Access: Protected

Headers: Authorization: Bearer <access_token>

Request Body:

```
{
  "firstName": "Ahmed",
  "lastName": "Mohamed",
  "phone": "+201009876543"
}
```

Success Response:

```
{
  "success": true,
  "message": "Profile updated successfully",
  "data": {
    "userId": 1,
    "firstName": "Ahmed",
    "lastName": "Mohamed",
    "email": "ahmed@example.com",
    "phone": "+201009876543",
    "role": "CUSTOMER",
    "userStatus": "ACTIVE"
  }
}
```

Failure Responses:

400 Bad Request

401 Unauthorized

3.5 Change Password Allows the logged-in user to change their password after providing the old password.

Method: PUT

Endpoint: /api/auth/change-password

Access: Protected

Headers: Authorization: Bearer <access_token>

Request Body:

```
{
  "oldPassword": "Password@123",
  "newPassword": "NewPassword@123"
}
```

Success Response:

```
{
  "success": true,
  "message": "Password changed successfully",
  "data": null
}
```

Failure Responses:

400 Invalid Old Password
401 Unauthorized

4. User/Profile APIs

4.1 Get All Customers Returns a paginated list of customer accounts in the system.

Method: GET

Endpoint: /api/users/customers

Access: SUPER_ADMIN

Query Parameters: page, size, status, keyword

Success Response:

```
{
  "success": true,
  "message": "Customers retrieved successfully",
  "data": {
    "content": [
      {
        "userId": 1,
        "firstName": "Ahmed",
        "lastName": "Ali",
        "email": "ahmed@example.com",
        "phone": "+201001234567",
        "role": "CUSTOMER",
        "userStatus": "ACTIVE"
      }
    ],
    "page": 0,
    "size": 10,
    "totalElements": 1,
  }
}
```

```
"totalPages": 1
}}
```

Failure Responses:

401 Unauthorized
403 Forbidden

4.2 Get Customer Details Returns the details of a specific customer account.

Method: GET

Endpoint: /api/users/customers/{customerId}

Access: SUPER_ADMIN

Path Parameter: customerId

Success Response:

```
{
  "success": true,
  "message": "Customer details retrieved successfully",
  "data": {
    "userId": 1,
    "firstName": "Ahmed",
    "lastName": "Ali",
    "email": "ahmed@example.com",
    "phone": "+201001234567",
    "role": "CUSTOMER",
    "userStatus": "ACTIVE"
  }
}
```

Failure Responses:

401 Unauthorized
403 Forbidden
404 Not Found

4.3 Deactivate Customer Account : Deactivates a customer account without deleting historical data.

Method: PATCH

Endpoint: /api/users/customers/{customerId}/deactivate

Access: SUPER_ADMIN

Path Parameter: customerId

Success Response:

```
{
  "success": true,
  "message": "Customer account deactivated successfully",
  "data": {
    "userId": 1,
    "email": "ahmed@example.com",
    "userStatus": "DEACTIVATED"
  }
}
```

Failure Responses:

401 Unauthorized

403 Forbidden

404 Not Found

409 Conflict

5. Airline & Admin APIs

5.1 Create Airline Creates a new airline in the system.

Method: POST

Endpoint: /api/airlines

Access: SUPER_ADMIN

Headers: Authorization: Bearer <access_token>

Request Body:

```
{
  "airlineName": "EgyptAir",
  "contactNumber": "+20212345678",
  "operatingRegion": "Middle East"
}
```

Success Response:

```
{
  "success": true,
  "message": "Airline created successfully",
  "data": {
    "airlineId": 1,
    "airlineName": "EgyptAir",
    "contactNumber": "+20212345678",
    "operatingRegion": "Middle East",
    "airlineStatus": "ACTIVE"
  }
}
```

Failure Responses:

400 Bad Request

401 Unauthorized

403 Forbidden

409 Conflict

5.2 Update Airline : Updates an existing airline information.

Method: PUT

Endpoint: /api/airlines/{airlineId}

Access: SUPER_ADMIN

Headers: Authorization: Bearer <access_token>

Path Parameter: airlineId

Request Body:

```
{
  "airlineName": "EgyptAir Updated",
  "contactNumber": "+20287654321",
  "operatingRegion": "Middle East and Africa"
}
```

Success Response:

```
{
  "success": true,
  "message": "Airline updated successfully",
  "data": {
    "airlineId": 1,
    "airlineName": "EgyptAir Updated",
    "contactNumber": "+20287654321",
    "operatingRegion": "Middle East and Africa",
    "airlineStatus": "ACTIVE"
  }
}
```

Failure Responses:

400 Bad Request

401 Unauthorized

403 Forbidden

404 Not Found

5.3 Deactivate Airline Deactivates an airline without deleting its historical flights, bookings, payments, or tickets.

Method: PATCH

Endpoint: /api/airlines/{airlineId}/deactivate

Access: SUPER_ADMIN.

Headers: Authorization: Bearer <access_token>

Path Parameter: airlineId

Success Response:

```
{
  "success": true,
  "message": "Airline deactivated successfully",
  "data": {
    "airlineId": 1,
    "airlineName": "EgyptAir",
    "airlineStatus": "INACTIVE"
  }
}
```

Failure Responses:

401 Unauthorized

403 Forbidden

404 Not Found

409 Conflict

5.4 Get All Airlines Returns a paginated list of airlines in the system.

Method: GET

Endpoint: /api/airlines

Access: SUPER_ADMIN

Headers: Authorization: Bearer <access_token>

Query Parameters: page, size, status, keyword

Success Response:

```
{
  "success": true,
  "message": "Airlines retrieved successfully",

```

```
"data": {
  "content": [
    {
      "airlineId": 1,
      "airlineName": "EgyptAir",
      "contactNumber": "+20212345678",
      "operatingRegion": "Middle East",
      "airlineStatus": "ACTIVE"
    }
  ],
  "page": 0,
  "size": 10,
  "totalElements": 1,
  "totalPages": 1
} }
```

Failure Responses:

401 Unauthorized

403 Forbidden

5.5 Create Airline Admin Creates a new Airline Admin account and assigns it to a specific airline.

Headers: Authorization: Bearer <access_token>

Method: POST

Endpoint: /api/airline-admins

Access: SUPER_ADMIN

Request Body:

```
{
  "airlineId": 1,
  "firstName": "Omar",
  "lastName": "Hassan",
  "email": "omar.admin@egyptair.com",
  "password": "Admin@123",
  "phone": "+201001112223"
}
```

Success Response:

```
{
  "success": true,
  "message": "Airline admin created successfully",
  "data": {
    "userId": 5,
    "airlineId": 1,
    "firstName": "Omar",
    "lastName": "Hassan",
    "email": "omar.admin@egyptair.com",
    "phone": "+201001112223",
    "role": "AIRLINE_ADMIN",
    "userStatus": "ACTIVE"
  }
}
```

Failure Responses:

400 Bad Request
401 Unauthorized
403 Forbidden
404 Not Found
409 Conflict

5.6 Update Airline Admin Updates an existing Airline Admin account.

Method: PUT

Endpoint: /api/airline-admins/{adminId}

Access: SUPER_ADMIN

Headers: Authorization: Bearer <access_token>

Path Parameter: adminId

Request Body:

```
{
  "airlineId": 1,
  "firstName": "Omar",
  "lastName": "Mostafa",
  "phone": "+201009999999"
}
```

Success Response:

```
{
  "success": true,
  "message": "Airline admin updated successfully",
  "data": {
    "userId": 5,
    "airlineId": 1,
    "firstName": "Omar",
    "lastName": "Mostafa",
    "email": "omar.admin@egyptair.com",
    "phone": "+201009999999",
    "role": "AIRLINE_ADMIN",
    "userStatus": "ACTIVE"
  }
}
```

Failure Responses:

400 Bad Request
401 Unauthorized
403 Forbidden
404 Not Found

5.7 Deactivate Airline Admin Deactivates an Airline Admin account without deleting historical records.

Method: PATCH

Endpoint: /api/airline-admins/{adminId}/deactivate

Access: SUPER_ADMIN

Headers: Authorization: Bearer <access_token>

Path Parameter: adminId

Success Response:

```
{
  "success": true,
  "message": "Airline admin deactivated successfully",
  "data": {
    "userId": 5,
    "email": "omar.admin@egyptair.com",
  }
}
```

```
"role": "AIRLINE_ADMIN",
"userStatus": "DEACTIVATED"
}
}
```

Failure Responses:

401 Unauthorized
403 Forbidden
404 Not Found
409 Conflict

5.8 Get All Airline Admins Returns a paginated list of Airline Admin accounts.

Method: GET

Endpoint: /api/airline-admins

Access: SUPER_ADMIN

Headers: Authorization: Bearer <access_token>

Query Parameters: page, size, status, airlineId, keyword

Success Response:

```
{
  "success": true,
  "message": "Airline admins retrieved successfully",
  "data": {
    "content": [
      {
        "userId": 5,
        "airlineId": 1,
        "airlineName": "EgyptAir",
        "firstName": "Omar",
        "lastName": "Hassan",
        "email": "omar.admin@egyptair.com",
        "phone": "+201001112223",
        "role": "AIRLINE_ADMIN",
        "userStatus": "ACTIVE"
      }
    ]
  }
}
```

```
],  
  "page": 0,  
  "size": 10,  
  "totalElements": 1,  
  "totalPages": 1  
}  
}
```

Failure Responses:

401 Unauthorized

403 Forbidden

6. Airport APIs Creates a new airport in the system.

6.1 Create Airport

Method: POST

Endpoint: /api/airports

Access: SUPER_ADMIN

Headers: Authorization: Bearer <access_token>

Request Body:

```
{  
  "airportCode": "CAI",  
  "airportName": "Cairo International Airport",  
  "city": "Cairo",  
  "country": "Egypt"  
}
```

Success Response:

```
{  
  "success": true,  
  "message": "Airport created successfully",  
  "data": {  
    "airportId": 1,  
    "airportCode": "CAI",  
    "airportName": "Cairo International Airport",  
  }  
}
```

```
"city": "Cairo",  
"country": "Egypt"  
}  
}
```

Failure Responses:

400 Bad Request
401 Unauthorized
403 Forbidden
409 Conflict

6.2 Update Airport Updates an existing airport information.

Method: PUT

Endpoint: /api/airports/{airportId}

Access: SUPER_ADMIN

Headers: Authorization: Bearer <access_token>

Path Parameter: airportId

Request Body:

```
{  
  "airportCode": "CAI",  
  "airportName": "Cairo International Airport",  
  "city": "Cairo",  
  "country": "Egypt"  
}
```

Success Response:

```
{  
  "success": true,  
  "message": "Airport updated successfully",  
  "data": {  
    "airportId": 1,  
    "airportCode": "CAI",  
    "airportName": "Cairo International Airport",  
  }  
}
```



```
"city": "Cairo",  
"country": "Egypt"  
}  
}
```

Failure Responses:

400 Bad Request
401 Unauthorized
403 Forbidden
404 Not Found
409 Conflict

6.3 Get All Airports Returns a paginated list of airports. This endpoint can be used by the frontend when searching for flights or selecting origin and destination airports.

Method: GET

Endpoint: /api/airports

Access: Public

Query Parameters: page, size, city, country, keyword

Success Response:

```
{  
  "success": true,  
  "message": "Airports retrieved successfully",  
  "data": {  
    "content": [  
      {  
        "airportId": 1,  
        "airportCode": "CAI",  
        "airportName": "Cairo International Airport",  
        "city": "Cairo",  
        "country": "Egypt"  
      }  
    ],  
    "page": 0,  
  }  
}
```

```
"size": 10,  
"totalElements": 1,  
"totalPages": 1  
}  
}
```

Failure Responses:

400 Bad Request

6.4 Get Airport Details Returns the details of a specific airport.

Method: GET

Endpoint: /api/airports/{airportId}

Access: Public

Path Parameter: airportId

Success Response:

```
{  
  "success": true,  
  "message": "Airport details retrieved successfully",  
  "data": {  
    "airportId": 1,  
    "airportCode": "CAI",  
    "airportName": "Cairo International Airport",  
    "city": "Cairo",  
    "country": "Egypt"  
  }  
}
```

Failure Responses:

404 Not Found

7. Flight APIs

7.1 Create Flight Creates a new flight for the airline assigned to the logged-in Airline Admin

Method: POST

Endpoint: /api/flights

Access: AIRLINE_ADMIN

Headers: Authorization: Bearer <access_token>

Request Body:

```
{ "flightNumber": "MS202",  
  "originAirportId": 1,  
  "destinationAirportId": 2,  
  "departureDateTime": "2026-07-10T10:30:00",  
  "arrivalDateTime": "2026-07-10T14:00:00",  
  "basePrice": 3500,  
  "seatCapacity": 180 }
```

Success Response:

```
{ "success": true,  
  "message": "Flight created successfully",  
  "data": {  
    "flightId": 10,  
    "airlineId": 1,  
    "flightNumber": "MS202",  
    "originAirportId": 1,  
    "destinationAirportId": 2,  
    "departureDateTime": "2026-07-10T10:30:00",  
    "arrivalDateTime": "2026-07-10T14:00:00",  
    "basePrice": 3500,  
    "seatCapacity": 180,  
    "flightStatus": "SCHEDULED"  
  } }
```

Failure Responses:

400 Bad Request
401 Unauthorized
403 Forbidden
404 Not Found
409 Conflict

7.2 Update Flight Updates flight information for a flight that belongs to the logged-in Airline Admin's assigned airline.

Method: PUT

Endpoint: /api/flights/{flightId}

Access: AIRLINE_ADMIN

Headers: Authorization: Bearer <access_token>

Path Parameter: flightId

Request Body:

```
{ "flightNumber": "MS202",  
  "originAirportId": 1,  
  "destinationAirportId": 2,  
  "departureDateTime": "2026-07-10T11:00:00",  
  "arrivalDateTime": "2026-07-10T14:30:00",  
  "basePrice": 3700,  
  "seatCapacity": 180 }
```

Success Response:

```
{ "success": true,  
  "message": "Flight updated successfully",  
  "data": {  
    "flightId": 10,  
    "airlineId": 1,  
    "flightNumber": "MS202",  
    "originAirportId": 1,  
    "destinationAirportId": 2,  
    "departureDateTime": "2026-07-10T11:00:00",  
    "arrivalDateTime": "2026-07-10T14:30:00",  
    "basePrice": 3700,  
    "seatCapacity": 180,  
    "flightStatus": "SCHEDULED"  
  } }
```

Failure Responses:

400 Bad Request
401 Unauthorized

403 Forbidden
404 Not Found

7.3 Cancel Flight Cancels a scheduled flight that belongs to the logged-in Airline Admin's assigned airline.

Method: PATCH

Endpoint: /api/flights/{flightId}/cancel

Access: AIRLINE_ADMIN

Headers: Authorization: Bearer <access_token>

Path Parameter: flightId

Success Response:

```
{
  "success": true,
  "message": "Flight cancelled successfully",
  "data": {
    "flightId": 10,
    "flightNumber": "MS202",
    "flightStatus": "CANCELLED"
  }
}
```

Failure Responses:

401 Unauthorized

403 Forbidden

404 Not Found

409 Conflict

7.4 Get All Flights Returns a paginated list of flights. Airline Admins can view only flights related to their assigned airline, while Super Admins can view all flights.

Method: GET

Endpoint: /api/flights

Access: AIRLINE_ADMIN, SUPER_ADMIN

Headers: Authorization: Bearer <access_token>

Query Parameters: page, size, airlineId, originAirportId, destinationAirportId, status, date, keyword

Success Response:

```
{
  "success": true,
  "message": "Flights retrieved successfully",
  "data": {
    "content": [
      {
        "flightId": 10,
        "airlineId": 1,
        "airlineName": "EgyptAir",
        "flightNumber": "MS202",
        "originAirportCode": "CAI",
        "destinationAirportCode": "DXB",
        "departureDateTime": "2026-07-10T10:30:00",
        "arrivalDateTime": "2026-07-10T14:00:00",
        "basePrice": 3500,
        "seatCapacity": 180,
        "flightStatus": "SCHEDULED"
      }
    ],
    "page": 0,
    "size": 10,
    "totalElements": 1,
    "totalPages": 1
  }
}
```

Failure Responses:

401 Unauthorized

403 Forbidden

7.5 Search Flights Searches for available flights based on origin, destination, departure date, passenger count, and availability.

Method: GET

Endpoint: /api/flights/search

Access: Public

Query Parameters: originAirportId, destinationAirportId, departureDate, passengerCount, page, size

Success Response:

```
{
  "success": true,
  "message": "Available flights retrieved successfully",
  "data": {
    "content": [
      {
        "flightId": 10,
        "airlineName": "EgyptAir",
        "flightNumber": "MS202",
        "originAirportCode": "CAI",
        "destinationAirportCode": "DXB",
        "departureDateTime": "2026-07-10T10:30:00",
        "arrivalDateTime": "2026-07-10T14:00:00",
        "basePrice": 3500,
        "availableSeats": 120,
        "flightStatus": "SCHEDULED"
      }
    ],
    "page": 0,
    "size": 10,
    "totalElements": 1,
    "totalPages": 1
  }
}
```

Failure Responses:

400 Bad Request

404 Not Found

7.6 Get Flight Details: Returns detailed information about a specific flight.

Method: GET

Endpoint: /api/flights/{flightId}

Access: Public

Path Parameter: flightId

Success Response:

```
{
  "success": true,
  "message": "Flight details retrieved successfully",
  "data": {
    "flightId": 10,
    "airlineId": 1,
    "airlineName": "EgyptAir",
    "flightNumber": "MS202",
    "originAirport": {
      "airportId": 1,
      "airportCode": "CAI",
      "airportName": "Cairo International Airport",
      "city": "Cairo",
      "country": "Egypt"
    },
    "destinationAirport": {
      "airportId": 2,
      "airportCode": "DXB",
      "airportName": "Dubai International Airport",
      "city": "Dubai",
      "country": "UAE"
    },
    "departureDateTime": "2026-07-10T10:30:00",
    "arrivalDateTime": "2026-07-10T14:00:00",
    "basePrice": 3500,
    "seatCapacity": 180,
    "availableSeats": 120,
    "flightStatus": "SCHEDULED"
  }
}
```

Failure Responses:

404 Not Found

8. Seat APIs Creates seats for a specific flight that belongs to the logged-in Airline Admin's assigned airline.

8.1 Create Seats for Flight

Method: POST

Endpoint: /api/flights/{flightId}/seats

Access: AIRLINE_ADMIN

Headers: Authorization: Bearer <access_token>

Path Parameter: flightId

Request Body:

```
{
  "seats": [
    {
      "seatNumber": "1A",
      "cabinClass": "ECONOMY"
    },
    {
      "seatNumber": "1B",
      "cabinClass": "ECONOMY"
    }
  ]
}
```

Success Response:

```
{
  "success": true,
  "message": "Seats created successfully",
  "data": {
    "flightId": 10,
    "createdSeats": 2
  }
}
```

Failure Responses:

400 Bad Request
401 Unauthorized
403 Forbidden
404 Not Found
409 Conflict

8.2 Update Seat Updates seat information such as seat number or cabin class.

Method: PUT

Endpoint: /api/seats/{seatId}

Access: AIRLINE_ADMIN

Headers: Authorization: Bearer <access_token>

Path Parameter: seatId

Request Body:

```
{
  "seatNumber": "2A",
  "cabinClass": "BUSINESS"
}
```

Success Response:

```
{
  "success": true,
  "message": "Seat updated successfully",
  "data": {
    "seatId": 15,
    "flightId": 10,
    "seatNumber": "2A",
    "cabinClass": "BUSINESS"
  }
}
```

Failure Responses:

400 Bad Request
401 Unauthorized
403 Forbidden
404 Not Found
409 Conflict

8.3 Get Flight Seats Returns all seats for a specific flight, including their cabin class and availability status.

Method: GET

Endpoint: /api/flights/{flightId}/seats

Access: Public

Path Parameter: flightId

Query Parameters: cabinClass, availability, page, size

Success Response:

```
{
  "success": true,
  "message": "Flight seats retrieved successfully",
  "data": {
    "content": [
      {
        "seatId": 15,
        "seatNumber": "2A",
        "cabinClass": "BUSINESS",
        "available": true
      },
      {
        "seatId": 16,
        "seatNumber": "2B",
        "cabinClass": "BUSINESS",
        "available": false
      }
    ],
    "page": 0,
    "size": 10,
    "totalElements": 2,
    "totalPages": 1
  }
}
```

Failure Responses:

400 Bad Request

404 Not Found

8.4 Get Available Seats for Flight Returns only available seats for a specific flight. This endpoint is used during the booking process before seat selection.

Method: GET

Endpoint: /api/flights/{flightId}/seats/available

Access: Public

Path Parameter: flightId

Query Parameters: cabinClass, page, size

Success Response:

```
{
  "success": true,
  "message": "Available seats retrieved successfully",
  "data": {
    "content": [
      {
        "seatId": 15,
        "seatNumber": "2A",
        "cabinClass": "BUSINESS"
      }
    ],
    "page": 0,
    "size": 10,
    "totalElements": 1,
    "totalPages": 1
  }
}
```

Failure Responses:

400 Bad Request

404 Not Found

9. Booking & Passenger APIs

9.1 Create Booking : Creates a new booking for a selected flight with PENDING_PAYMENT status.

Method: POST

Endpoint: /api/bookings

Access: CUSTOMER

Headers: Authorization: Bearer <access_token>

Request Body:

```
{
  "flightId": 10,
  "passengerCount": 2
}
```

Success Response:

```
{
  "success": true,
  "message": "Booking created successfully",
  "data": {
    "bookingId": 1001,
    "bookingReference": "FB-2026-0001",
    "customerId": 1,
    "flightId": 10,
    "bookingStatus": "PENDING_PAYMENT",
    "totalAmount": 7000
  }
}
```

Failure Responses:

400 Bad Request
401 Unauthorized
403 Forbidden
404 Not Found
409 Conflict

9.2 Add Passengers and Select Seats : Adds passenger details to a booking and assigns a seat to each passenger.

Method: POST

Endpoint: /api/bookings/{bookingId}/passengers

Access: CUSTOMER

Headers: Authorization: Bearer <access_token>

Path Parameter: bookingId

Request Body:

```
{ "passengers": [  
  {  
    "firstName": "Ahmed",  
    "lastName": "Ali",  
    "passportNumber": "A1234567",  
    "nationality": "Egyptian",  
    "seatId": 15  
  }, {  
    "firstName": "Sara",  
    "lastName": "Ali",  
    "passportNumber": "B9876543",  
    "nationality": "Egyptian",  
    "seatId": 16  
  } ] }
```

Success Response:

```
{ "success": true,  
  "message": "Passengers added successfully",  
  "data": {  
    "bookingId": 1001,  
    "passengerCount": 2,  
    "bookingStatus": "PENDING_PAYMENT"  
  } }
```

Failure Responses:

400 Bad Request

401 Unauthorized

403 Forbidden

404 Not Found

409 Conflict

9.3 Get My Bookings : Returns a paginated list of bookings for the logged-in customer.

Method: GET

Endpoint: /api/bookings/my

Access: CUSTOM

Headers: Authorization: Bearer <access_token>

Query Parameters: page, size, status

Success Response:

```
{
  "success": true,
  "message": "Bookings retrieved successfully",
  "data": {
    "content": [
      {
        "bookingId": 1001,
        "bookingReference": "FB-2026-0001",
        "flightNumber": "MS202",
        "bookingStatus": "PENDING_PAYMENT",
        "totalAmount": 7000
      }
    ],
    "page": 0,
    "size": 10,
    "totalElements": 1,
    "totalPages": 1
  }
}
```

Failure Responses:

401 Unauthorized

403 Forbidden

9.4 Get Booking Details : Returns full details of a specific booking, including flight data, passengers, seats, and booking status.

Method: GET

Endpoint: /api/bookings/{bookingId}

Access: CUSTOMER, AIRLINE_ADMIN, SUPER_ADMIN

Headers: Authorization: Bearer <access_token>

Path Parameter: bookingId

Success Response:

```
{
  "success": true,
  "message": "Booking details retrieved successfully",
  "data": {
    "bookingId": 1001,
    "bookingReference": "FB-2026-0001",
    "bookingStatus": "PENDING_PAYMENT",
    "totalAmount": 7000,
    "flight": {
      "flightId": 10,
      "flightNumber": "MS202",
      "originAirportCode": "CAI",
      "destinationAirportCode": "DXB"
    },
    "passengers": [
      {
        "passengerId": 1,
        "firstName": "Ahmed",
        "lastName": "Ali",
        "passportNumber": "A1234567",
        "nationality": "Egyptian",
        "seatNumber": "2A"
      }
    ]
  }
}
```

Failure Responses:

401 Unauthorized
403 Forbidden
404 Not Found

9.5 Cancel Booking Cancels an eligible booking. If the booking is already paid and confirmed, refund eligibility will be checked later in the Refund APIs.

Method: PATCH

Endpoint: /api/bookings/{bookingId}/cancel

Access: CUSTOMER

Headers: Authorization: Bearer <access_token>

Path Parameter: bookingId

Success Response:

```
{
  "success": true,
  "message": "Booking cancelled successfully",
  "data": {
    "bookingId": 1001,
    "bookingReference": "FB-2026-0001",
    "bookingStatus": "CANCELLED"
  }
}
```

Failure Responses:

401 Unauthorized
403 Forbidden
404 Not Found
409 Conflict

9.6 Get Airline Bookings Returns bookings related only to the logged-in Airline Admin's assigned airline.

Method: GET

Endpoint: /api/bookings/airline

Access: AIRLINE_ADMIN

Headers: Authorization: Bearer <access_token>

Query Parameters: page, size, status, flightId

Success Response:

```
{
  "success": true,
  "message": "Airline bookings retrieved successfully",
  "data": {
    "content": [
      {
        "bookingId": 1001,
        "bookingReference": "FB-2026-0001",
        "customerName": "Ahmed Ali",
        "flightNumber": "MS202",

```

```
        "bookingStatus": "CONFIRMED",
        "totalAmount": 7000
    }
],
"page": 0,
"size": 10,
"totalElements": 1,
"totalPages": 1
}
}
```

Failure Responses:

401 Unauthorized

403 Forbidden

9.7 Get All Bookings Returns a paginated list of all bookings across the system.

Method: GET

Endpoint: /api/bookings

Access: SUPER_ADMIN

Headers: Authorization: Bearer <access_token>

Query Parameters: page, size, status, airlineId, flightId, customerId

Success Response:

```
{
  "success": true,
  "message": "All bookings retrieved successfully",
  "data": {
    "content": [
      {
        "bookingId": 1001,
        "bookingReference": "FB-2026-0001",
        "customerName": "Ahmed Ali",
        "airlineName": "EgyptAir",
        "flightNumber": "MS202",

```

```
      "bookingStatus": "CONFIRMED",  
      "totalAmount": 7000  
    }  
  ],  
  "page": 0,  
  "size": 10,  
  "totalElements": 1,  
  "totalPages": 1  
}  
}
```

Failure Responses:

401 Unauthorized

403 Forbidden

10. Payment APIs

10.1 Create Payment Request : Creates a payment request for a booking with PENDING_PAYMENT status.

Method: POST

Endpoint: /api/payments

Access: CUSTOMER

Headers: Authorization: Bearer <access_token>

Request Body:

```
{
  "bookingId": 1001,
  "paymentMethod": "CARD"
}
```

Success Response:

```
{
  "success": true,
  "message": "Payment request created successfully",
  "data": {
    "paymentId": 501,
    "bookingId": 1001,
    "amount": 7000,
    "paymentMethod": "CARD",
    "paymentStatus": "PENDING",
    "transactionReference": "TXN-2026-0001"
  }
}
```

Failure Responses:

400 Bad Request

401 Unauthorized

403 Forbidden

404 Not Found

409 Conflict

10.2 Get Payment Details : Returns details of a specific payment.

Method: GET

Endpoint: /api/payments/{paymentId}

Access: CUSTOMER, SUPER_ADMIN

Headers: Authorization: Bearer <access_token>

Path Parameter: paymentId

Success Response:

```
{
  "success": true,
  "message": "Payment details retrieved successfully",
  "data": {
    "paymentId": 501,
    "bookingId": 1001,
    "amount": 7000,
    "paymentMethod": "CARD",
    "transactionReference": "TXN-2026-0001",
    "transactionDateTime": "2026-07-02T18:30:00",
    "paymentStatus": "SUCCESS"
  }
}
```

Failure Responses:

401 Unauthorized

403 Forbidden

404 Not Found

10.3 Get Booking Payment : Returns the payment information related to a specific booking.

Method: GET

Endpoint: /api/bookings/{bookingId}/payment

Access: CUSTOMER, AIRLINE_ADMIN, SUPER_ADMIN

Headers: Authorization: Bearer <access_token>

Path Parameter: bookingId

Success Response:

```
{
  "success": true,
  "message": "Booking payment retrieved successfully",
  "data": {
    "paymentId": 501,
    "bookingId": 1001,
    "amount": 7000,
    "paymentMethod": "CARD",
    "transactionReference": "TXN-2026-0001",
  }
}
```

```
    "paymentStatus": "SUCCESS"
  }
}
```

Failure Responses:

401 Unauthorized

403 Forbidden

404 Not Found

10.4 Payment Callback Receives the payment result from the Payment Gateway and updates the payment status. If the payment is successful, the booking status is updated to CONFIRMED.

Method: POST

Endpoint: /api/payments/callback

Access: PAYMENT_GATEWAY

Headers: X-Gateway-Signature: <gateway_signature>

Request Body:

```
{
  "transactionReference": "TXN-2026-0001",
  "paymentStatus": "SUCCESS",
  "amount": 7000,
  "paidAt": "2026-07-02T18:30:00"
}
```

Success Response:

```
{
  "success": true,
  "message": "Payment result processed successfully",
  "data": {
    "paymentId": 501,
    "bookingId": 1001,
    "paymentStatus": "SUCCESS",
    "bookingStatus": "CONFIRMED"
  }
}
```

Failure Responses:

400 Bad Request

401 Unauthorized

404 Not Found

409 Conflict

10.5 Get My Payments : Returns a paginated list of payments for the logged-in customer.

Method: GET

Endpoint: /api/payments/my

Access: CUSTOMER

Headers: Authorization: Bearer <access_token>

Query Parameters: page, size, status

Success Response:

```
{
  "success": true,
  "message": "Payments retrieved successfully",
  "data": {
    "content": [
      {
        "paymentId": 501,
        "bookingId": 1001,
        "amount": 7000,
        "paymentMethod": "CARD",
        "paymentStatus": "SUCCESS",
        "transactionReference": "TXN-2026-0001"
      }
    ],
    "page": 0,
    "size": 10,
    "totalElements": 1,
    "totalPages": 1
  }
}
```

Failure Responses:

401 Unauthorized

403 Forbidden

11. Refund APIs

11.1 Request Refund : Creates a refund request for an eligible paid and cancelled booking.

Method: POST

Endpoint: /api/refunds

Access: CUSTOMER

Headers: Authorization: Bearer <access_token>

Request Body:

```
{
  "bookingId": 1001,
  "reason": "Customer cancelled the booking"
}
```

Success Response:

```
{
  "success": true,
  "message": "Refund request created successfully",
  "data": {
    "refundId": 301,
    "paymentId": 501,
    "bookingId": 1001,
    "refundAmount": 7000,
    "refundStatus": "PENDING",
    "refundReference": "REF-2026-0001"
  }
}
```

Failure Responses:

400 Bad Request

401 Unauthorized

403 Forbidden

404 Not Found

409 Conflict

422 Unprocessable Entity

11.2 Get Refund Details : Returns the details of a specific refund.

Method: GET

Endpoint: /api/refunds/{refundId}

Access: CUSTOMER, AIRLINE_ADMIN, SUPER_ADMIN

Headers: Authorization: Bearer <access_token>

Path Parameter: refundId

Success Response:

```
{
  "success": true,
  "message": "Refund details retrieved successfully",
  "data": {
    "refundId": 301,
    "paymentId": 501,
    "bookingId": 1001,
    "refundAmount": 7000,
    "refundStatus": "REFUNDED",
    "refundReference": "REF-2026-0001",
    "requestedAt": "2026-07-02T19:00:00",
    "processedAt": "2026-07-02T19:10:00",
    "failureReason": null
  }
}
```

Failure Responses:

401 Unauthorized
Returned when the JWT token is missing, invalid, or expired.

403 Forbidden
Returned when the user does not have permission to view this refund.

404 Not Found
Returned when the refund does not exist.

11.3 Get Booking Refund : Returns the refund information related to a specific booking.

Method: GET

Endpoint: /api/bookings/{bookingId}/refund

Access: CUSTOMER, AIRLINE_ADMIN, SUPER_ADMIN

Headers: Authorization: Bearer <access_token>

Path Parameter: bookingId

Success Response:

```
{
  "success": true,
  "message": "Booking refund retrieved successfully",
  "data": {
    "refundId": 301,
    "bookingId": 1001,
    "paymentId": 501,
  }
}
```

```
"refundAmount": 7000,  
"refundStatus": "PENDING",  
"refundReference": "REF-2026-0001"  
}  
}
```

Failure Responses:

401 Unauthorized
403 Forbidden
404 Not Found

11.4 Refund Callback : Receives the refund result from the Payment Gateway and updates the refund status.

Method: POST

Endpoint: /api/refunds/callback

Access: PAYMENT_GATEWAY

Headers: X-Gateway-Signature: <gateway_signature>

Request Body:

```
{  
  "refundReference": "REF-2026-0001",  
  "refundStatus": "REFUNDED",  
  "refundAmount": 7000,  
  "processedAt": "2026-07-02T19:10:00",  
  "failureReason": null  
}
```

Success Response:

```
{  
  "success": true,  
  "message": "Refund result processed successfully",  
  "data": {  
    "refundId": 301,  
    "paymentId": 501,  
    "refundStatus": "REFUNDED"  
  }  
}
```

Failure Responses:

400 Bad Request
401 Unauthorized
404 Not Found
409 Conflict

11.5 Get My Refunds : Returns a paginated list of refunds requested by the logged-in customer.

Method: GET

Endpoint: /api/refunds/my

Access: CUSTOMER

Headers: Authorization: Bearer <access_token>

Query Parameters: page, size, status

Success Response:

```
{
  "success": true,
  "message": "Refunds retrieved successfully",
  "data": {
    "content": [
      {
        "refundId": 301,
        "bookingId": 1001,
        "paymentId": 501,
        "refundAmount": 7000,
        "refundStatus": "REFUNDED",
        "refundReference": "REF-2026-0001"
      }
    ],
    "page": 0,
    "size": 10,
    "totalElements": 1,
    "totalPages": 1
  }
}
```

Failure Responses:

401 Unauthorized

403 Forbidden

11.6 Get All Refunds: Returns a paginated list of all refunds across the system.

Method: GET

Endpoint: /api/refunds

Access: SUPER_ADMIN

Headers: Authorization: Bearer <access_token>

Query Parameters: page, size, status, airlineId, bookingId, paymentId

Success Response:

```
{
  "success": true,
  "message": "All refunds retrieved successfully",
  "data": {
    "content": [
      {
        "refundId": 301,
        "bookingId": 1001,
        "paymentId": 501,
        "customerName": "Ahmed Ali",
        "airlineName": "EgyptAir",
        "refundAmount": 7000,
        "refundStatus": "REFUNDED",
        "refundReference": "REF-2026-0001"
      }
    ],
    "page": 0,
    "size": 10,
    "totalElements": 1,
    "totalPages": 1
  }
}
```

Failure Responses:

401 Unauthorized
403 Forbidden

12. Ticket APIs

12.1 Generate Tickets for Booking : Generates tickets for all passengers in a confirmed booking. This endpoint is triggered internally after successful payment confirmation.

Method: POST

Endpoint: /api/bookings/{bookingId}/tickets/generate

Access: SYSTEM_INTERNAL

Headers: Authorization: Bearer <internal_token>

Path Parameter: bookingId

Success Response:

```
{
  "success": true,
  "message": "Tickets generated successfully",
  "data": {
    "bookingId": 1001,
    "bookingReference": "FB-2026-0001",
    "generatedTickets": 2,
    "ticketStatus": "ISSUED"
  }
}
```

Failure Responses:

401 Unauthorized
404 Not Found
409 Conflict
422 Unprocessable Entity

12.2 Get My Tickets : Returns a paginated list of tickets for the logged-in customer.

Method: GET

Endpoint: /api/tickets/my

Access: CUSTOMER

Headers: Authorization: Bearer <access_token>

Query Parameters: page, size, status

Success Response:

```
{
  "success": true,
  "message": "Tickets retrieved successfully",
  "data": {
    "content": [
      {
        "ticketId": 701,
        "ticketNumber": "TKT-2026-0001",
        "bookingId": 1001,
        "bookingReference": "FB-2026-0001",
        "passengerName": "Ahmed Ali",
        "flightNumber": "MS202",
        "ticketStatus": "ISSUED",
        "issuedAt": "2026-07-02T18:35:00"
      }
    ],
    "page": 0,
    "size": 10,
    "totalElements": 1,
    "totalPages": 1
  }
}
```

Failure Responses:

401 Unauthorized

Returned when the JWT token is missing, invalid, or expired.

403 Forbidden

Returned when the user is not a Customer.

12.3 Get Ticket Details : Returns full details of a specific ticket.

Method: GET

Endpoint: /api/tickets/{ticketId}

Access: CUSTOMER, AIRLINE_ADMIN, SUPER_ADMIN

Headers: Authorization: Bearer <access_token>

Path Parameter: ticketId

Success Response:

```
{
  "success": true,
  "message": "Ticket details retrieved successfully",
  "data": {
    "ticketId": 701,
    "ticketNumber": "TKT-2026-0001",
```

```

"ticketStatus": "ISSUED",
"issuedAt": "2026-07-02T18:35:00",
"booking": {
  "bookingId": 1001,
  "bookingReference": "FB-2026-0001",
  "bookingStatus": "CONFIRMED"
},
"passenger": {
  "passengerId": 1,
  "firstName": "Ahmed",
  "lastName": "Ali",
  "passportNumber": "A1234567",
  "nationality": "Egyptian",
  "seatNumber": "2A"
},
"flight": {
  "flightId": 10,
  "flightNumber": "MS202",
  "originAirportCode": "CAI",
  "destinationAirportCode": "DXB",
  "departureDateTime": "2026-07-10T10:30:00",
  "arrivalDateTime": "2026-07-10T14:00:00"
}
}
}

```

Failure Responses:

401 Unauthorized
 403 Forbidden
 404 Not Found

12.4 Get Booking Tickets : Returns all tickets related to a specific booking.

Method: GET

Endpoint: /api/bookings/{bookingId}/tickets

Access: CUSTOMER, AIRLINE_ADMIN, SUPER_ADMIN

Headers: Authorization: Bearer <access_token>

Path Parameter: bookingId

Success Response:

```

{
  "success": true,
  "message": "Booking tickets retrieved successfully",
  "data": [
    {
      "ticketId": 701,

```

```
{
  "ticketNumber": "TKT-2026-0001",
  "passengerName": "Ahmed Ali",
  "seatNumber": "2A",
  "ticketStatus": "ISSUED",
  "issuedAt": "2026-07-02T18:35:00"
},
{
  "ticketId": 702,
  "ticketNumber": "TKT-2026-0002",
  "passengerName": "Sara Ali",
  "seatNumber": "2B",
  "ticketStatus": "ISSUED",
  "issuedAt": "2026-07-02T18:35:00"
}
]
```

Failure Responses:

401 Unauthorized
403 Forbidden
404 Not Found

12.5 Cancel Ticket: Cancels or invalidates a ticket when the related booking is cancelled.

Method: PATCH

Endpoint: /api/tickets/{ticketId}/cancel

Access: SYSTEM_INTERNAL, SUPER_ADMIN

Headers: Authorization: Bearer <access_token>

Path Parameter: ticketId

Success Response:

```
{ "success": true,
  "message": "Ticket cancelled successfully",
  "data": {
    "ticketId": 701,
    "ticketNumber": "TKT-2026-0001",
    "ticketStatus": "CANCELLED"
  }
}
```

Failure Responses:

401 Unauthorized
403 Forbidden
404 Not Found
409 Conflict

13. Notification APIs

13.1 Send Notification : Sends a notification email to a user. This endpoint is triggered internally by the system after booking confirmation, ticket generation, cancellation, refund processing, or flight update.

Method: POST

Endpoint: /api/notifications/send

Access: SYSTEM_INTERNAL

Headers: Authorization: Bearer <internal_token>

Request Body:

```
{
  "userId": 1,
  "bookingId": 1001,
  "notificationType": "BOOKING_CONFIRMATION",
  "recipientEmail": "ahmed@example.com",
  "subject": "Booking Confirmation",
  "message": "Your booking has been confirmed successfully."
}
```

Success Response:

```
{
  "success": true,
  "message": "Notification sent successfully",
  "data": {
    "notificationId": 901,
    "notificationType": "BOOKING_CONFIRMATION",
    "notificationStatus": "SENT",
    "sentAt": "2026-07-03T15:30:00"
  }
}
```

Failure Responses:

400 Bad Request

401 Unauthorized

404 Not Found

500 Internal Server Error

13.2 Get My Notifications : Returns a paginated list of notifications for the logged-in customer.

Method: GET

Endpoint: /api/notifications/my

Access:

Headers: Authorization: Bearer <access_token>

Query Parameters: page, size, type, status

Success Response:

```
{
  "success": true,
  "message": "Notifications retrieved successfully",
  "data": {
    "content": [
      {
        "notificationId": 901,
        "bookingId": 1001,
        "notificationType": "BOOKING_CONFIRMATION",
        "notificationStatus": "SENT",
        "subject": "Booking Confirmation",
        "sentAt": "2026-07-03T15:30:00"
      }
    ],
    "page": 0,
    "size": 10,
    "totalElements": 1,
    "totalPages": 1
  }
}
```

Failure Responses:

401 Unauthorized

403 Forbidden

13.3 Get Notification Details : Returns the details of a specific notification.

Method: GET

Endpoint: /api/notifications/{notificationId}

Access: CUSTOMER, SUPER_ADMIN

Headers: Authorization: Bearer <access_token>

Path Parameter: notificationId

Success Response:

```
{
  "success": true,
  "message": "Notification details retrieved successfully",
  "data": {
    "notificationId": 901,
    "userId": 1,
    "bookingId": 1001,
    "notificationType": "BOOKING_CONFIRMATION",
    "notificationStatus": "SENT",
    "recipientEmail": "ahmed@example.com",
    "subject": "Booking Confirmation",
    "message": "Your booking has been confirmed successfully.",
    "sentAt": "2026-07-03T15:30:00",
    "failureReason": null
  }
}
```

Failure Responses:

401 Unauthorized

403 Forbidden

404 Not Found

13.4 Get All Notifications: Returns a paginated list of all notifications in the system.

Method: GET

Endpoint: /api/notifications

Access: SUPER_ADMIN

Headers: Authorization: Bearer <access_token>

Query Parameters: page, size, type, status, userId, bookingId

Success Response:

```
{
  "success": true,
  "message": "All notifications retrieved successfully",
  "data": {
    "content": [
      {
        "notificationId": 901,
        "userId": 1,
        "bookingId": 1001,
        "notificationType": "BOOKING_CONFIRMATION",
        "notificationStatus": "SENT",
        "recipientEmail": "ahmed@example.com",
        "subject": "Booking Confirmation",
        "sentAt": "2026-07-03T15:30:00"
      }
    ],
    "page": 0,
    "size": 10,
    "totalElements": 1,
    "totalPages": 1
  }
}
```

Failure Responses:

401 Unauthorized
403 Forbidden

13.5 Retry Failed Notification: Retries sending a failed notification.

Method: POST

Endpoint: /api/notifications/{notificationId}/retry

Access: SUPER_ADMIN

Headers: Authorization: Bearer <access_token>

Path Parameter: notificationId

Success Response:

```
{  
  "success": true,  
  "message": "Notification retry processed successfully",  
  "data": {  
    "notificationId": 901,  
    "notificationStatus": "SENT",  
    "sentAt": "2026-07-03T15:40:00",  
    "failureReason": null  
  }  
}
```

Failure Responses:

401 Unauthorized
403 Forbidden
404 Not Found
409 Conflict

14. Reports & Audit APIs

14.1 Get Booking Report : Returns booking report data. Super Admin can view all bookings, while Airline Admin can view only bookings related to their assigned airline.

Method: GET

Endpoint: /api/reports/bookings

Access: AIRLINE_ADMIN, SUPER_ADMIN

Headers: Authorization: Bearer <access_token>

Query Parameters: fromDate, toDate, airlineId, flightId, status, page, size

Success Response:

```
{
  "success": true,
  "message": "Booking report retrieved successfully",
  "data": {
    "totalBookings": 120,
    "confirmedBookings": 90,
    "cancelledBookings": 20,
    "pendingPaymentBookings": 10,
    "totalRevenue": 250000,
    "content": [
      {
        "bookingId": 1001,
        "bookingReference": "FB-2026-0001",
        "customerName": "Ahmed Ali",
        "airlineName": "EgyptAir",
        "flightNumber": "MS202",
        "bookingStatus": "CONFIRMED",
        "totalAmount": 7000
      }
    ],
    "page": 0,
    "size": 10,
    "totalElements": 120,
    "totalPages": 12
  }
}
```

Failure Responses:

400 Bad Request
401 Unauthorized
403 Forbidden

14.2 Get Payment Report : Returns payment report data across the system.

Method: GET

Endpoint: /api/reports/payments

Access: SUPER_ADMIN

Headers: Authorization: Bearer <access_token>

Query Parameters: fromDate, toDate, status, paymentMethod, page, size

Success Response:

```
{
  "success": true,
  "message": "Payment report retrieved successfully",
  "data": {
    "totalPayments": 100,
    "successfulPayments": 85,
    "failedPayments": 15,
    "totalPaidAmount": 230000,
    "content": [
      {
        "paymentId": 501,
        "bookingId": 1001,
        "amount": 7000,
        "paymentMethod": "CARD",
        "paymentStatus": "SUCCESS",
        "transactionReference": "TXN-2026-0001"
      }
    ],
    "page": 0,
    "size": 10,
    "totalElements": 100,
    "totalPages": 10  } }
```

Failure Responses:

400 Bad Request
401 Unauthorized
403 Forbidden

14.3 Get Refund Report : Returns refund report data across the system.

Method: GET

Endpoint: /api/reports/refunds

Access: SUPER_ADMIN

Headers: Authorization: Bearer <access_token>

Query Parameters: fromDate, toDate, status, airlineId, page, size

Success Response:

```
{
  "success": true,
  "message": "Refund report retrieved successfully",
  "data": {
    "totalRefunds": 25,
    "refundedCount": 20,
    "failedRefunds": 5,
    "totalRefundedAmount": 50000,
    "content": [
      {
        "refundId": 301,
        "paymentId": 501,
        "bookingId": 1001,
        "refundAmount": 7000,
        "refundStatus": "REFUNDED",
        "refundReference": "REF-2026-0001"
      }
    ],
    "page": 0,
    "size": 10,
    "totalElements": 25,
    "totalPages": 3
  }
}
```

Failure Responses:

400 Bad Request

401 Unauthorized

403 Forbidden

14.4 Get Audit Logs: Returns a paginated list of audit logs for sensitive system actions.

Method: GET

Endpoint: /api/audit-logs

Access: SUPER_ADMIN

Headers: Authorization: Bearer <access_token>

Query Parameters: page, size, userId, action, entityName, fromDate, toDate

Success Response:

```
{
  "success": true,
  "message": "Audit logs retrieved successfully",
  "data": {
    "content": [
      {
        "auditLogId": 1,
        "userId": 10,
        "action": "CREATE_AIRLINE_ADMIN",
        "entityName": "USERS",
        "entityId": 5,
        "details": "Super Admin created a new Airline Admin account.",
        "createdAt": "2026-07-03T16:00:00"
      }
    ],
    "page": 0,
    "size": 10,
    "totalElements": 1,
    "totalPages": 1
  }
}
```

Failure Responses:

400 Bad Request

401 Unauthorized

403 Forbidden

14.5 Get Audit Log Details: Returns details of a specific audit log record.

Method: GET

Endpoint: /api/audit-logs/{auditLogId}

Access: SUPER_ADMIN

Headers: Authorization: Bearer <access_token>

Path Parameter: auditLogId

Success Response:

```
{ "success": true,  
  "message": "Audit log details retrieved successfully",  
  "data": {  
    "auditLogId": 1,  
    "userId": 10,  
    "action": "CREATE_AIRLINE_ADMIN",  
    "entityName": "USERS",  
    "entityId": 5,  
    "details": "Super Admin created a new Airline Admin account.",  
    "createdAt": "2026-07-03T16:00:00" } }
```

Failure Responses:

401 Unauthorized

403 Forbidden

404 Not Found
